

MATH 227B: Mathematical Biology

Homework 4

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March 5, 2026

Code availability

All the code used for these problems is available publicly at this [GitHub repository](#).

Problem 1

a) For the following scheme,

$$y_{n+1}^* = y_n + \frac{h}{2} (3f(y_n) - f(y_{n-1})),$$
$$y_{n+1} = y_n + \frac{h}{2} (f(y_n) + f(y_{n+1}^*)),$$

(a) Derive the local truncation error.

(b) Derive and plot the regions of absolute stability. Next, implement both schemes on computer. Solve the following ODEs using both schemes:

$$\frac{dy_1}{dt} = -5y_1 + 3y_2,$$
$$\frac{dy_2}{dt} = 100y_1 - 301y_2,$$

with initial conditions $y_1(0) = 52.29$ and $y_2(0) = 83.82$.

(c) Is the accuracy of the solutions in the scheme, when solving the above system on computer, consistent with your analysis of local truncation error?

(d) Discuss your choice of h when solving the above equations.

Local Truncation Error of the Scheme

Consider the initial value problem $y'(t) = f(y(t))$ and the predictor–corrector (PECE) numerical scheme:

$$\begin{aligned} y_{n+1}^* &= y_n + \frac{h}{2} (3f(y_n) - f(y_{n-1})), \\ y_{n+1} &= y_n + \frac{h}{2} (f(y_n) + f(y_{n+1}^*)). \end{aligned}$$

We define the local truncation error (LTE) as $\tau_{n+1} = y(t_{n+1}) - y_{n+1}$, assuming y_n and y_{n-1} are exact.

Step 1: Taylor Expansions of the Exact Solution

The exact solution $y(t_{n+1})$ and the derivative $f(y(t_{n-1}))$ expanded about t_n are

$$y(t_{n+1}) = y_n + hf_n + \frac{h^2}{2}f'_n + \frac{h^3}{6}f''_n + O(h^4), \quad (1)$$

$$f(y_{n-1}) = f_n - hf'_n + \frac{h^2}{2}f''_n + O(h^3), \quad (2)$$

where $f_n = f(y(t_n))$, $f'_n = y''(t_n)$, and $f''_n = y'''(t_n)$.

Predictor LTE (Adams–Bashforth 2-step)

Substituting (2) into the predictor formula gives

$$\begin{aligned} y_{n+1}^* &= y_n + \frac{h}{2} \left[3f_n - \left(f_n - hf'_n + \frac{h^2}{2}f''_n + O(h^3) \right) \right] \\ &= y_n + \frac{h}{2} \left(2f_n + hf'_n - \frac{h^2}{2}f''_n + O(h^3) \right) \\ &= y_n + hf_n + \frac{h^2}{2}f'_n - \frac{h^3}{4}f''_n + O(h^4). \end{aligned}$$

Comparing this with (1) we obtain

$$\tau_p = y(t_{n+1}) - y_{n+1}^* = \left(\frac{1}{6} - \left(-\frac{1}{4} \right) \right) h^3 f''_n + O(h^4) = \frac{5}{12} h^3 y'''_n + O(h^4).$$

Thus

Predictor LTE: $\tau_p = \frac{5}{12} h^3 y'''_n + O(h^4)$ (second order globally).

Corrector LTE (Theoretical Trapezoidal Rule)

If the corrector were using the exact value $y(t_{n+1})$ in the function evaluation, we would have

$$\begin{aligned} y_{n+1}^{\text{corr}} &= y_n + \frac{h}{2} [f_n + f(y(t_{n+1}))] \\ &= y_n + \frac{h}{2} \left[f_n + \left(f_n + hf'_n + \frac{h^2}{2} f''_n + O(h^3) \right) \right] \\ &= y_n + hf_n + \frac{h^2}{2} f'_n + \frac{h^3}{4} f''_n + O(h^4). \end{aligned}$$

Comparing with (1) yields

$$\tau_c = y(t_{n+1}) - y_{n+1}^{\text{corr}} = \left(\frac{1}{6} - \frac{1}{4} \right) h^3 f''_n + O(h^4) = -\frac{1}{12} h^3 y'''_n + O(h^4),$$

and therefore

Corrector LTE: $\tau_c = -\frac{1}{12} h^3 y'''_n + O(h^4)$ (second order globally).

Predictor–Corrector LTE (Full PECE Scheme)

In the full scheme, we evaluate $f(y_{n+1}^*)$. Since $y_{n+1}^* = y(t_{n+1}) + O(h^3)$, a Taylor expansion of f around $y(t_{n+1})$ implies

$$f(y_{n+1}^*) = f(y(t_{n+1})) + f_y(y(t_{n+1})) O(h^3) = f(y(t_{n+1})) + O(h^3).$$

The full update can thus be written as

$$\begin{aligned} y_{n+1} &= y_n + \frac{h}{2} (f_n + f(y_{n+1}^*)) \\ &= y_n + \frac{h}{2} (f_n + f(y(t_{n+1})) + O(h^3)) \\ &= y_{n+1}^{\text{corr}} + O(h^4), \end{aligned}$$

because the $O(h^3)$ discrepancy in $f(y_{n+1}^*)$ is multiplied by an extra factor of h . Therefore the leading term of the local error remains that of the trapezoidal rule:

Full PC LTE: $\tau_{pc} = -\frac{1}{12} h^3 y'''_n + O(h^4)$ (second order globally).

Summary of Error Scalings

The PECE scheme inherits the $O(h^3)$ local truncation error of the trapezoidal rule and is therefore a second-order global method.

Method	Local Error (LTE)	Global Error (GTE)
Predictor (AB2)	$O(h^3)$	$O(h^2)$
Corrector (AM2)	$O(h^3)$	$O(h^2)$
Full PC scheme	$O(h^3)$	$O(h^2)$

Table 1: Local and global error orders for the predictor, corrector and full PECE scheme.

Regions of Absolute Stability

To analyze stability, we apply the method to the linear test equation

$$y' = \lambda y, \quad \lambda \in C.$$

Let

$$z = h\lambda.$$

Since $f(y) = \lambda y$, the numerical scheme becomes a linear recurrence relation in y_n .

Stability of the Predictor

The predictor step is

$$\begin{aligned} y_{n+1}^* &= y_n + \frac{h}{2}(3\lambda y_n - \lambda y_{n-1}) \\ &= y_n + \frac{z}{2}(3y_n - y_{n-1}). \end{aligned} \tag{3}$$

Thus

$$y_{n+1}^* = \left(1 + \frac{3}{2}z\right) y_n - \frac{z}{2} y_{n-1}.$$

To analyze stability we assume solutions of the form

$$y_n = \xi^n.$$

Substituting into the recurrence relation yields

$$\xi^{n+1} = \left(1 + \frac{3}{2}z\right) \xi^n - \frac{z}{2} \xi^{n-1}.$$

Dividing by ξ^{n-1} gives the characteristic polynomial

$$\xi^2 - \left(1 + \frac{3}{2}z\right) \xi + \frac{z}{2} = 0.$$

The predictor is absolutely stable for those values of z such that

$$|\xi_1(z)| \leq 1, \quad |\xi_2(z)| \leq 1,$$

where ξ_1, ξ_2 are the roots of this polynomial.

Stability of the Predictor (AB2)

From (3), applied to the test equation $y' = \lambda y$ with $z = h\lambda$, we see that the stability polynomial of the AB2 predictor is

$$\xi^2 - \left(1 + \frac{3}{2}z\right)\xi + \frac{z}{2} = 0.$$

The region of absolute stability $\mathcal{S}_{\text{pred}}$ is the set of $z \in \mathbb{C}$ such that all roots ξ satisfy $|\xi| \leq 1$. For AB2 this is a bounded region contained entirely in the left half-plane, with a real-axis intercept approximately on $(-1, 0)$; thus the predictor alone is rather restrictive for stiff problems.

Stability of the Corrector (AM2)

The corrector step, if solved implicitly to convergence (the trapezoidal rule), is

$$y_{n+1} = y_n + \frac{z}{2}(y_{n+1} + y_n).$$

Rearranging for the amplification factor $R(z) = y_{n+1}/y_n$ gives

$$y_{n+1} \left(1 - \frac{z}{2}\right) = y_n \left(1 + \frac{z}{2}\right) \implies R(z) = \frac{1 + z/2}{1 - z/2}.$$

The trapezoidal rule is A -stable, meaning $|R(z)| \leq 1$ for all $\text{Re}(z) \leq 0$, corresponding to the entire left-half complex plane.

Stability of the Predictor–Corrector Scheme (PECE)

In the full PECE implementation, the predicted value y_{n+1}^* is substituted into the corrector. Using $y_{n+1}^* = y_n + \frac{z}{2}(3y_n - y_{n-1})$ from the predictor, we obtain

$$\begin{aligned} y_{n+1} &= y_n + \frac{z}{2}(y_n + y_{n+1}^*) \\ &= y_n + \frac{z}{2}\left(y_n + \left[y_n + \frac{z}{2}(3y_n - y_{n-1})\right]\right) \\ &= y_n + \frac{z}{2}\left(2y_n + \frac{3z}{2}y_n - \frac{z}{2}y_{n-1}\right). \end{aligned}$$

Expanding and collecting terms yields

$$y_{n+1} = \left(1 + z + \frac{3}{4}z^2\right)y_n - \frac{z^2}{4}y_{n-1}.$$

Substituting $y_n = \xi^n$ gives the characteristic polynomial for the full PC scheme:

$$\xi^2 - \left(1 + z + \frac{3}{4}z^2\right)\xi + \frac{1}{4}z^2 = 0.$$

This scheme is absolutely stable when both roots $\xi_{1,2}(z)$ satisfy $|\xi| \leq 1$.

Summary of Stability Criteria

The three regions analyzed can be summarized as

$$\begin{aligned}\mathcal{S}_{\text{pred}} &= \left\{ z \in C : \xi^2 - \left(1 + \frac{3}{2}z\right)\xi + \frac{z}{2} = 0, |\xi| \leq 1 \right\}, \\ \mathcal{S}_{\text{corr}} &= \{z \in C : \text{Re}(z) \leq 0\} \quad (\text{implicit trapezoidal rule}), \\ \mathcal{S}_{\text{PC}} &= \left\{ z \in C : \xi^2 - \left(1 + z + \frac{3}{4}z^2\right)\xi + \frac{1}{4}z^2 = 0, |\xi| \leq 1 \right\}.\end{aligned}$$

Numerical Stability Analysis and Visual Interpretation

The stability characteristics of the three methods are visualized in Figure 5. By mapping the amplification factors across the complex z -plane (where $z = h\lambda$), we can categorize the robustness of each approach for the given stiff system.

From the numerical plots the AB2 predictor has a real-axis intercept near $z \approx -1$, whereas the PECE scheme extends this to roughly $z \approx -1.25$, while the AM2 corrector covers the entire left half-plane, consistent with the analytical characterization above (see also the axis annotations in the PDF stability figures).

Solver Implementation

The numerical solution of linear initial value problems was implemented using two closely related schemes: the explicit two-step Adams–Bashforth (AB2) predictor and the predictor–corrector scheme employing a trapezoidal corrector (PECE). The **predictor-only scheme** advances the solution using stored derivatives from previous time steps, providing a second-order global approximation when used as a standalone AB2 method. The **predictor–corrector scheme** improves accuracy by correcting the predicted solution using the trapezoidal rule, resulting in second-order global accuracy with a higher constant of accuracy and improved stability properties.

To initiate the two-step methods, a standard **Runge–Kutta 4 (RK4) step** is used for the first time step, ensuring the necessary starting value with high accuracy. At each subsequent step, the predictor computes a provisional solution, which is then corrected by evaluating the derivative at this predicted state. This correction ensures that the derivative used in the next step is consistent with the updated solution. These schemes are particularly suitable for linear systems $y' = Ay$ but generalize to nonlinear problems as well. A portion of the implementation code is shown in Listing 1.

Listing 1: Python code for predictor and predictor–corrector IVP solvers

```
1 import numpy as np
2
3 def f_linear(t, y, A):
4     return A @ y
```

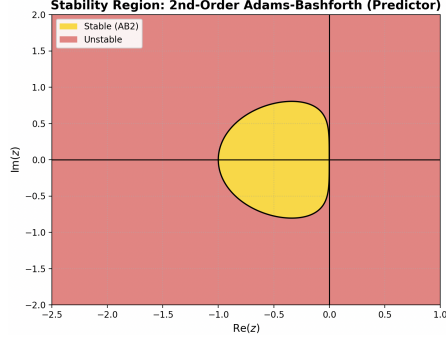


Figure 1: Predictor (AB2). The shaded region indicates values of z for which both characteristic roots satisfy $|\xi| \leq 1$, forming a bounded lobe in the left half-plane.

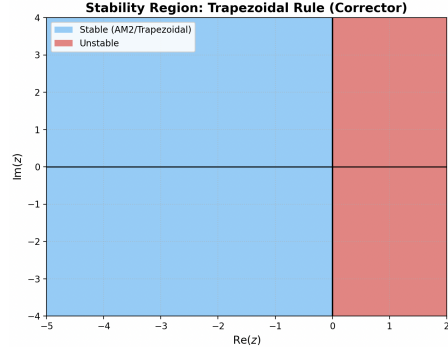


Figure 2: Corrector (AM2). The trapezoidal rule is A -stable, with stability extending over the entire left half-plane in the z -plane.

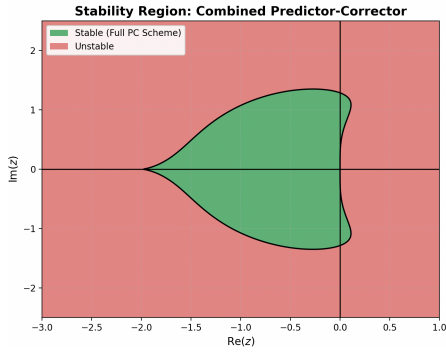


Figure 3: Combined PECE scheme. The stability region strictly contains that of AB2 and remains a bounded subset of the left half-plane.

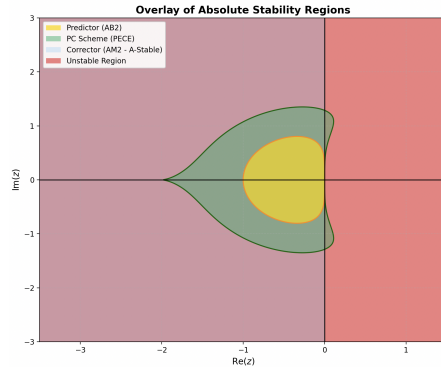


Figure 4: Overlay of stability regions. The nesting $\mathcal{S}_{\text{pred}} \subset \mathcal{S}_{\text{PC}} \subset \mathcal{S}_{\text{corr}}$ is evident from the figure.

Figure 5: Regions of absolute stability in the complex z -plane. Shaded areas denote the stable regions ($|\xi| \leq 1$). Note the progression of stability from the explicit AB2 to the A -stable implicit AM2, with the implemented PECE scheme providing an intermediate enlargement of the explicit stability domain.

```

5
6 def predictor_corrector_step(y_n, f_n, f_nm1, h, f, t_np1, *
  f_args):
7     y_star = y_n + 0.5*h*(3.0*f_n - f_nm1)  # Predictor (AB2
      )
8     f_star = f(t_np1, y_star, *f_args)      # Derivative at
      predicted state
9     y_np1 = y_n + 0.5*h*(f_n + f_star)      # Corrector (
      trapezoidal)
10    f_np1 = f(t_np1, y_np1, *f_args)        # Derivative at
      corrected state
11    return y_np1, f_np1, y_star
12
13 def initialize_two_step_rk4(y0, t0, h, f, *f_args):
14     k1 = f(t0, y0, *f_args)
15     k2 = f(t0 + 0.5*h, y0 + 0.5*h*k1, *f_args)
16     k3 = f(t0 + 0.5*h, y0 + 0.5*h*k2, *f_args)
17     k4 = f(t0 + h, y0 + h*k3, *f_args)
18     return y0 + (h/6)*(k1 + 2*k2 + 2*k3 + k4)
19
20 def solve_predictor_corrector(f, t_span, y0, h, *f_args):
21     t0, T = t_span
22     y0 = np.atleast_1d(y0)
23     N = int((T-t0)/h)
24     t = t0 + h*np.arange(N+1)
25     Y = np.zeros((N+1, y0.size))
26     F = np.zeros((N+1, y0.size))
27     Y[0] = y0
28     F[0] = f(t[0], Y[0], *f_args)
29     if N > 0:
30         Y[1] = initialize_two_step_rk4(Y[0], t[0], h, f, *
          f_args)
31         F[1] = f(t[1], Y[1], *f_args)
32     for n in range(1, N):
33         Y[n+1], F[n+1], _ = predictor_corrector_step(
34             Y[n], F[n], F[n-1], h, f, t[n+1], *f_args
35         )
36     return t, Y

```

Unit Tests of the Solver

To verify the correctness of the predictor–corrector solver implementation, we applied it to a suite of benchmark ODE systems with known analytical solutions. For each system, the numerical solution was computed at every time step and compared against the exact solution using the condition

$$\|y_{\text{numerical}}(t_n) - y_{\text{exact}}(t_n)\| < \text{tolerance} \quad \forall n,$$

ensuring that the solver produces accurate results at all intermediate points in addition to the final time.

The systems tested included a variety of scalar, oscillatory and coupled linear problems, designed to probe different aspects of solver performance (stability, handling of complex eigenvalues, sequential dependencies and stiffness). The table below summarizes the systems, initial conditions and the results of these unit tests.

- **F1: Scalar exponential decay** — $y' = -ky$, $y(0) = y_0$, exact solution $y(t) = y_0 e^{-kt}$.
- **F2: Simple harmonic oscillator** — $y'_1 = y_2$, $y'_2 = -y_1$, exact solution $y_1(t) = \cos(t)$, $y_2(t) = -\sin(t)$.
- **F3: Coupled linear decay** — $y' = Ay$ with a non-stiff matrix A having real negative eigenvalues.
- **F4: Circular orbit system** — Hamiltonian system $y'_1 = -y_2$, $y'_2 = y_1$, conserving $y_1^2 + y_2^2$.
- **F5: Sequential decay chain** — Triangular matrix with dependent dynamics.
- **F6: Stable spiral** — Complex eigenvalues with negative real part, simulating damped oscillations.
- **F7: Stiff-like linear system** — Matrix with widely separated eigenvalues to test robustness.

ID	Equation / dynamics	Initial condition	Test result
F1	$y' = -ky$	$y_0 = 1.0$	Pass
F2	$y'_1 = y_2$, $y'_2 = -y_1$	$y_0 = [1.0, 0.0]$	Pass
F3	$y' = Ay$	$y_0 = [10.0, 5.0]$	Pass
F4	$y'_1 = -y_2$, $y'_2 = y_1$	$y_0 = [1.0, 0.0]$	Pass
F5	Triangular system	$y_0 = [1.0, 0.0]$	Pass
F6	$y' = Ay$, complex eigenvalues	$y_0 = [1.0, 0.0]$	Pass
F7	$y' = Ay$, widely separated eigenvalues	$y_0 = [1.0, 1.0]$	Pass

Table 2: Summary of unit tests for the predictor–corrector solver. Numerical solutions were verified at every time step against exact solutions. All tests passed, confirming the correctness of the implementation across diverse systems.

In summary, the predictor–corrector solver passed all unit tests across the range of benchmark problems, confirming that the numerical implementation correctly handles scalar decay, oscillatory dynamics, coupled linear systems, sequential chains, damped spirals and stiff-like behavior.

Visual Verification of the Solver

To complement the automated unit tests, we performed a series of visual validations of the numerical solvers. The objective was to confirm that the predictor-only (AB2) and predictor-corrector (PECE) schemes reproduce the expected qualitative and quantitative behavior of representative benchmark systems. For this analysis, we selected two systems that are particularly informative:

1. **Simple harmonic oscillator (SHO):** $y'_1 = y_2$, $y'_2 = -y_1$, with exact solution $y_1(t) = \cos(t)$, $y_2(t) = -\sin(t)$. This system tests the ability of the solver to preserve amplitude and phase over multiple oscillation periods, highlighting both stability and accuracy in oscillatory dynamics.
2. **Coupled linear decay:** $y' = Ay$ with $A = \begin{bmatrix} -2 & 0.5 \\ 0.1 & -1.5 \end{bmatrix}$, initial condition $y(0) = [10, 5]^T$. This system captures interactions between coupled variables with differing decay rates, testing the solver's handling of dependencies and mild stiffness effects.

For each system, the following visual validation procedure was applied:

- Numerical trajectories were computed using both the predictor-only and predictor-corrector solvers for a range of five timestep values h . The trajectories were plotted against the exact solution to visually assess amplitude, phase, and overall fidelity.
- Global truncation error (GTE) was calculated at the final time for each timestep h and plotted on a log-log scale to verify the expected convergence rates: $O(h^2)$ for the AB2 predictor and $O(h^2)$ (with a smaller error constant) for the PECE scheme.
- Local truncation error (LTE) was similarly assessed over a single step for each h , confirming the theoretical $O(h^3)$ behavior for both schemes.

Validation for the harmonic oscillator. Figure 6 shows the time trajectories for the SHO system under both solver schemes. Across all tested timesteps, the predictor-corrector solver closely tracks the exact solution, preserving the amplitude and phase of the oscillations over multiple periods. In contrast, the predictor-only scheme shows noticeable phase drift and amplitude distortion for larger timesteps, consistent with its more restrictive stability domain. The error-versus- h plots confirm slopes close to 2 in the log-log GTE curves for both methods, with the PECE scheme yielding uniformly smaller errors for a given timestep, as evidenced by the separated lines in the bottom panels of the figure.

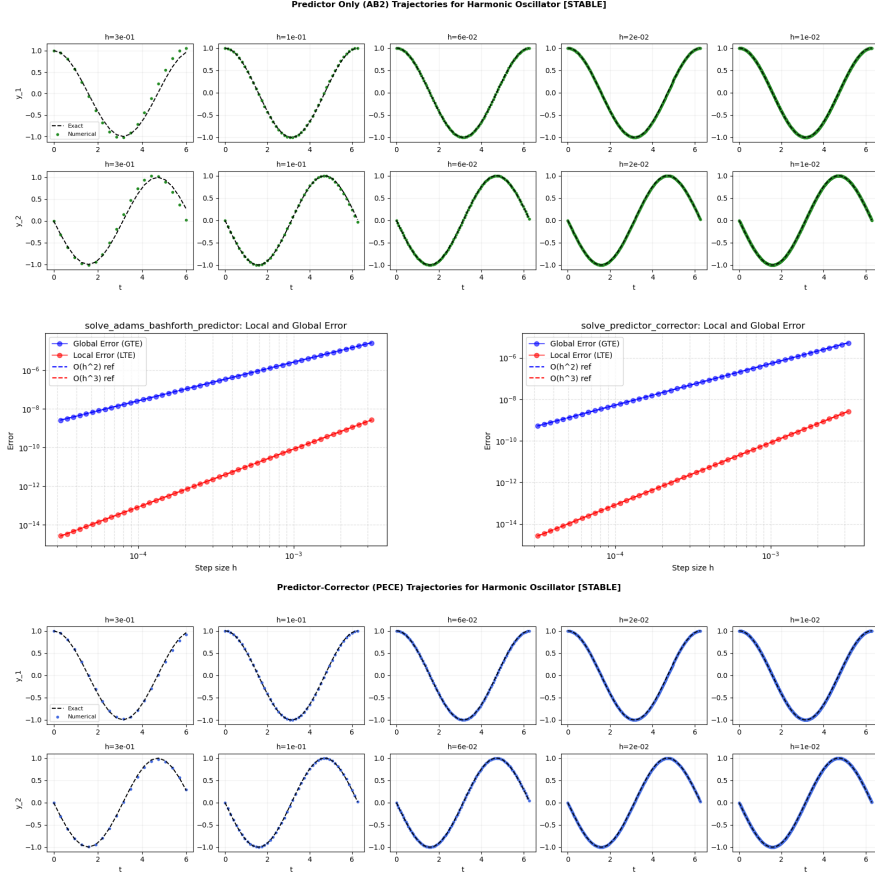


Figure 6: Validation of the predictor-only (AB2) and predictor-corrector (PECE) schemes for the **harmonic oscillator (F2)**. Top: numerical trajectories for five timestep values h overlaid with the exact solution, demonstrating accurate preservation of amplitude and phase as h decreases. Middle: global error versus time for AB2 (left) and PECE (right). Bottom: final-time global error and maximum local truncation error versus h on a log-log scale, confirming approximately second-order convergence for both schemes and uniformly smaller errors for PECE.

Validation for the coupled linear decay. Figure 7 displays the numerical trajectories for the coupled decay system. Both variables, y_1 and y_2 , follow the exact solution closely with the predictor-corrector solver across all timesteps. The predictor-only solver exhibits increasing deviation for larger h , particularly in the slower-decaying component y_2 , highlighting the sensitivity of coupled dynamics to cumulative phase and amplitude errors. The GTE and LTE curves again exhibit slopes consistent with second-order global and third-order local

accuracy, with PECE producing noticeably smaller absolute errors for a given h .

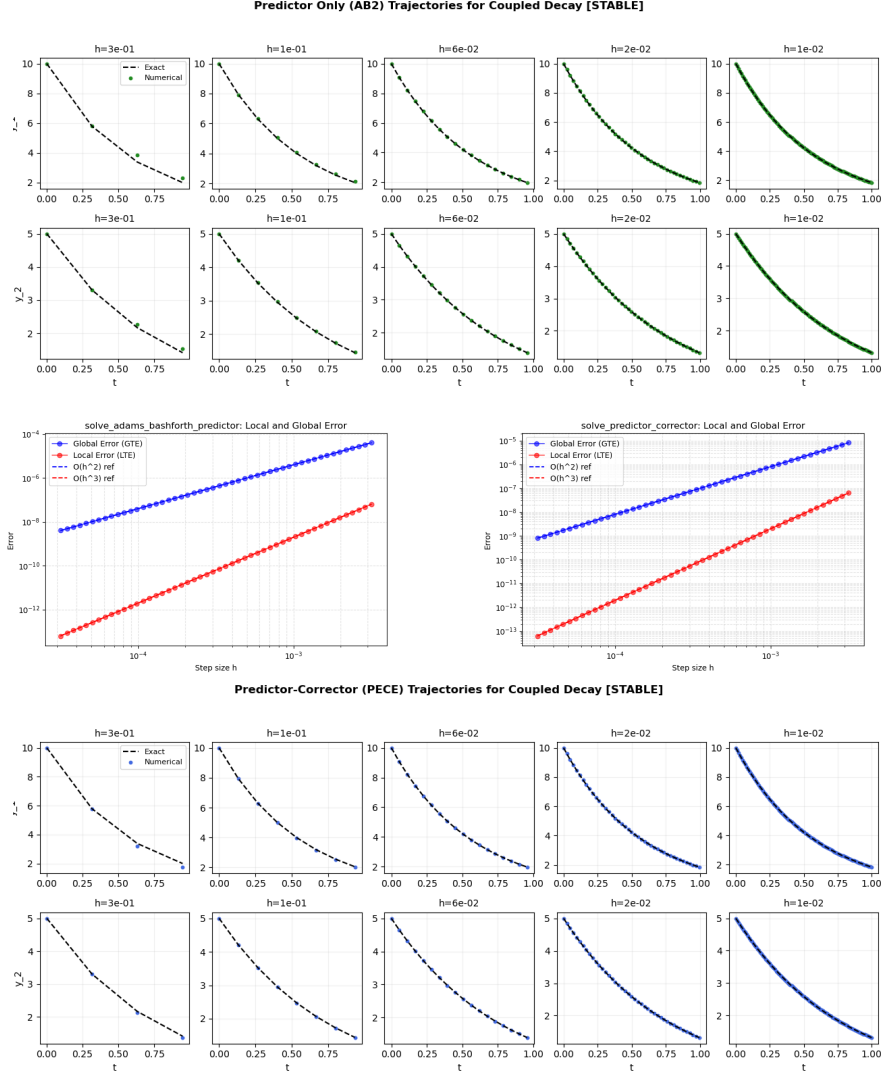


Figure 7: Validation of the predictor-only (AB2) and predictor-corrector (PECE) schemes for the **coupled decay system (F3)**. Top: numerical trajectories for five timestep values h overlaid with the exact solution. Middle: global error versus time for AB2 (left) and PECE (right). Bottom: final-time global error and maximum local truncation error versus h on a log-log scale, showing second-order convergence and systematically smaller errors for the PECE scheme.

Observations. These visual analyses reinforce the unit test results and demonstrate that:

- The predictor–corrector scheme consistently achieves higher accuracy and preserves qualitative dynamics across a range of timestep sizes.
- The predictor-only scheme exhibits error growth consistent with its theoretical order, with noticeable phase or amplitude deviations in oscillatory or coupled systems for coarser timesteps.
- Both global and local error plots versus h match theoretical expectations, providing strong evidence that the solvers are implemented correctly.

Overall, combining trajectory plots with GTE and LTE analyses provides a comprehensive visual verification framework that complements automated unit tests.

Applying the Solver to the Given System

We now apply both the predictor-only (AB2) and predictor–corrector (PECE) schemes to the linear system

$$\frac{dy_1}{dt} = -5y_1 + 3y_2, \quad (4)$$

$$\frac{dy_2}{dt} = 100y_1 - 301y_2, \quad (5)$$

with initial conditions $y_1(0) = 52.29$, $y_2(0) = 83.82$. The system can be written compactly as

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y}, \quad A = \begin{bmatrix} -5 & 3 \\ 100 & -301 \end{bmatrix}.$$

Eigenvalue analysis of A yields

$$\lambda_1 \approx -1, \quad \lambda_2 \approx -305,$$

highlighting the *stiff-like* character of the system due to the widely separated timescales. The fast mode (λ_2) decays rapidly, whereas the slow mode (λ_1) governs the long-term evolution.

Numerical setup. Integration was performed over the interval $t \in [0, 1]$ using both AB2 and PECE schemes. Step sizes h were chosen to probe stability and convergence, ranging from values well inside to near the boundary of the stability region. To systematically assess solver behavior, we computed:

- Time trajectories $y_1(t), y_2(t)$ for several step sizes h and compared them to the exact solution obtained from the matrix exponential.
- Global truncation error (GTE) at the final time $T = 1$ for each h .

- Local truncation error (LTE) at each time step for each h , visualized as a heatmap on the (h, t) -plane.

Time trajectories and error analysis. Figure 8 summarizes the results. For five representative step sizes h , the top panels show the numerical trajectories of $y_1(t)$ and $y_2(t)$ using AB2 and PECE overlaid with the exact solution. Panels (c) and (d) display the final-time GTE versus h on a log-log scale, confirming approximately second-order convergence for both schemes. Panels (e) and (f) present log-scaled heatmaps of LTE as a function of h and t , revealing vertical bands of larger error near the initial transient associated with the fast eigenmode ($\lambda_2 \approx -305$) and systematic decay of LTE with decreasing h .

The trajectory plots reveal that the predictor-only scheme suffers from noticeable overshoot and oscillations when h is too large, especially for $y_2(t)$, whereas the predictor-corrector scheme maintains accurate amplitude and phase even for moderately large h . The error-versus- h curves confirm that the PECE scheme achieves smaller global errors for each step size, in agreement with the local truncation error analysis and the larger effective stability region observed in Figure 5.

Step Size Selection and Stability Analysis

The stiff-like nature of the assignment system, characterized by widely separated eigenvalues $\lambda_1 \approx -1$ and $\lambda_2 \approx -305$, imposes strict constraints on timestep selection. Explicit methods, such as the AB2 predictor, are particularly sensitive to the fast mode associated with λ_2 , whereas the predictor-corrector scheme exhibits enhanced stability. To systematically assess the effect of the timestep h on numerical behavior, we selected three representative values based on the stability regions previously computed for each method. These stability regions, obtained by analyzing the magnitude of the method amplification factor $R(z)$ in the complex plane, yield the condition

$$|R(h\lambda_i)| \leq 1, \quad i = 1, 2,$$

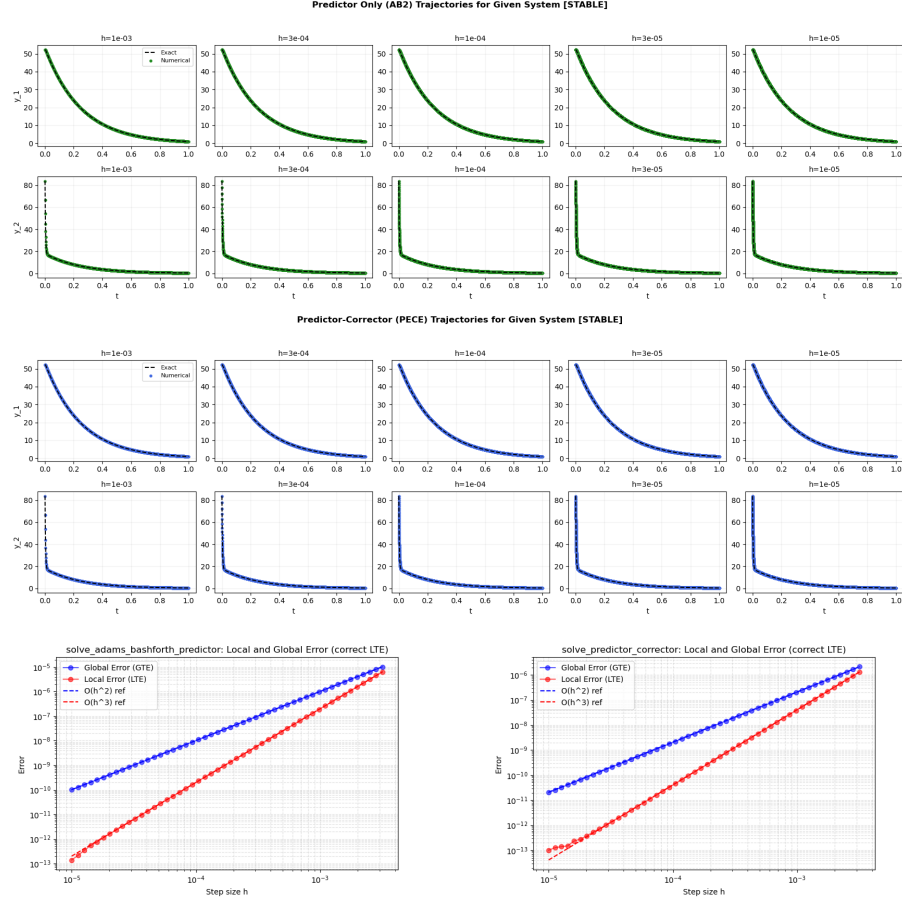
where $R(z)$ corresponds to the predictor or predictor-corrector scheme and λ_i is an eigenvalue of the system matrix.

The selected step sizes are:

$$h = 0.001 \quad (\text{stable for both AB2 and PECE}), \quad h = 0.004 \quad (\text{stable for PECE but near the AB2 stability limit})$$

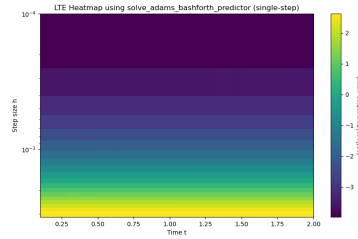
These values were chosen by identifying representative points along the real axis in the stability regions and solving for h such that $h\lambda_2$ lies just inside or outside the corresponding boundary.

Figure 9 illustrates the numerical consequences of these choices. Trajectories for $y_1(t)$ and $y_2(t)$ are plotted on a logarithmic scale to clearly highlight the onset of divergence. Red vertical lines indicate the approximate time at which the predictor-only solution begins to deviate rapidly from the exact solution for

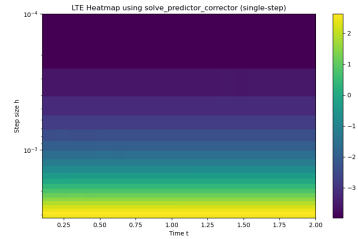


AB2 error analysis (log-log).

PECE error analysis (log-log).



AB2 LTE heatmap in (h, t) .



PECE LTE heatmap.

Figure 8: Trajectories and error analysis of the given stiff-like system for AB2 and PECE schemes over five step sizes. The trajectory plots highlight overshoot and oscillations for overly large h with AB2, while PECE maintains accurate amplitude and phase for moderately larger steps.

each unstable h , emphasizing the effect of instability in the explicit scheme. In contrast, the PECE trajectories remain bounded for intermediate h and diverge only for the largest timestep, consistent with the predicted stability regions.

In addition to trajectory plots, we analyzed the global and local errors as functions of step size. The bottom panels of Figure 9 compare AB2 and PECE: both the final-time GTE and the maximum LTE across all timesteps increase rapidly as h approaches the stability limits. For step sizes where AB2 becomes unstable, the errors diverge sharply, whereas PECE maintains bounded errors until the largest tested h , corroborating the superior stability of the predictor–corrector method for stiff-like systems.

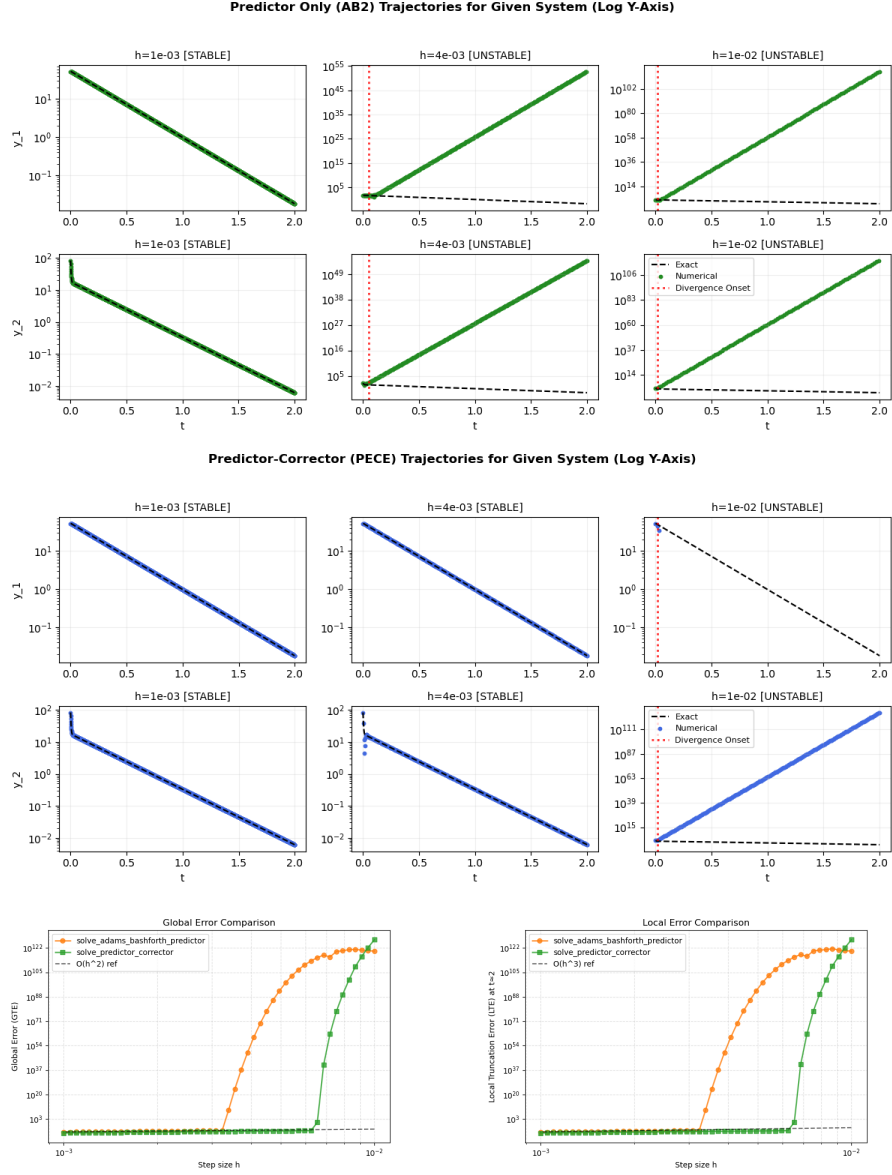
Observations and Discussion

The results reveal several key aspects of solver behavior for the stiff-like assignment system. First, the large negative eigenvalue λ_2 strongly constrains the timestep for explicit methods: if h is too large, the predictor scheme exhibits oscillatory divergence that is clearly visible in the log-scaled trajectories. Second, the predictor–corrector scheme is more robust, allowing moderately larger timesteps while still accurately capturing both the fast and slow modes. Third, both global and local errors escalate sharply as h approaches the stability limit, confirming the theoretical predictions from the local truncation error and stability analysis. Finally, the combination of trajectory, GTE and LTE plots provides a comprehensive picture of both quantitative and qualitative solver performance, supporting informed timestep selection that balances accuracy, stability and computational efficiency.

Across all test systems and the assignment problem, we observe consistent patterns. The AB2 predictor exhibits second-order convergence in global error with a relatively restrictive stability region, while the predictor–corrector scheme achieves the same formal global order but with smaller error constants and an enlarged stability domain. Visual verification through trajectories and LTE heatmaps highlights the impact of stiff eigenmodes and transient dynamics. Step size selection emerges as a critical factor, particularly for systems with widely separated eigenvalues: small timesteps are required to resolve fast modes accurately; intermediate timesteps exploit the stability advantage of PECE; and large timesteps lead to numerical blow-up. These observations underline the importance of combining analytical, quantitative and visual assessments when validating numerical solvers for stiff or oscillatory systems.

Problem 2

1) Reproduce the simulation results in Figure 1 and 2 of the following paper: [Liu, Xinfeng, et al. “Nonlinear growth kinetics of breast cancer stem cells: implications for cancer stem cell targeted therapy.” *Scientific Reports* 3.1 \(2013\): 2473.](#)



Global error comparison (log-log)

Maximum LTE comparison (log-log)

Figure 9: Numerical validation of the given system using AB2 (top left) and PECE (top right) schemes across selected step sizes. Trajectories are plotted on a logarithmic scale with red vertical lines indicating the approximate onset of divergence for unstable runs. Bottom panels show the corresponding global error and maximum local truncation error as functions of h .

2) **Extra Credits:** Use the model in the paper or its variation to fit (or study) one additional set of published experimental data on tumor growth not given in this paper.

Cancer stem cells (CSCs) are believed to drive tumor growth and recurrence through their ability to self-renew and differentiate into multiple tumor cell types. The study by Liu et al. [2] proposes a nonlinear mathematical model for tumor cell population dynamics that incorporates delayed negative feedback from differentiated tumor cells. The model explains experimentally observed fluctuations in the fraction of CSCs during tumor cell culture.

A central challenge in reproducing the published results is that the *original paper does not specify the initial conditions used in the simulations*. Because delay differential equation (DDE) models are sensitive to initial states and history functions, careful reasoning is required to infer plausible initial conditions from the experimental context. This report therefore develops a systematic strategy for reconstructing the missing initial conditions and evaluates the resulting simulations. The reproduction process proceeds in three stages:

1. Implement the mathematical models described in the paper.
2. Attempt to reproduce the published figures using reported parameter values with inferred initial values.
3. Extract the experimental data points from the published figures and perform parameter fitting to evaluate how well the model can match the observations.

Mathematical Model of Tumor Cell Population Dynamics

Cell population structure. The model describes tumor growth using three cellular compartments: cancer stem cells (CSCs), progenitor cancer cells (PCs) and terminally differentiated cells (TDCs). CSCs can self-renew or differentiate, producing progenitor cells. Progenitor cells proliferate for a limited number of generations before differentiating into terminally differentiated cells.

Feedback regulation. The central assumption of Liu et al. [2] is that CSC dynamics are regulated by negative feedback from the differentiated cell compartment. Biologically, progenitor and terminally differentiated cells produce inhibitory signals that reduce CSC self-renewal and proliferation. This feedback is delayed due to the time required for differentiation and signaling, which is captured in a system of delay differential equations:

$$\frac{d}{dt} \begin{pmatrix} x_0(t) \\ x_1(t) \\ x_2(t) \end{pmatrix} = F \left(x_0(t), x_1(t), x_2(t), x_2(t - \tau) \right),$$

where x_0 , x_1 and x_2 denote CSCs, progenitor cells and terminally differentiated cells, respectively, and τ represents the feedback delay.

Type I feedback: proliferation modulation. In the Type I feedback model, differentiated cells reduce the *proliferation rates* of CSCs and progenitors. Mathematically, the baseline proliferation rates v_i are scaled by a decreasing function of the delayed differentiated cell number:

$$v_i^{\text{eff}}(x_2(t-\tau)) = \frac{v_i}{1 + \beta_i x_2(t-\tau)^2}, \quad i = 0, 1,$$

where $\beta_i > 0$ controls feedback strength. This ensures that higher numbers of differentiated cells slow the expansion of stem and progenitor compartments.

Type II feedback: self-renewal modulation. Type II feedback targets the *self-renewal and symmetric division probabilities* rather than proliferation rates. Effective division probabilities p_i^{eff} and q_i^{eff} decrease with the delayed differentiated cell population:

$$p_i^{\text{eff}}(x_2(t-\tau)) = \frac{p_i}{1 + \gamma_{i0} x_2(t-\tau)^2}, \quad q_i^{\text{eff}}(x_2(t-\tau)) = \frac{q_i}{1 + \gamma_{i1} x_2(t-\tau)^2}, \quad i = 0, 1,$$

with $\gamma_{ij} > 0$ controlling the feedback strength. This reduces the probability of symmetric self-renewal events when differentiated cells accumulate.

Combined feedback. The combined model integrates both mechanisms: proliferation rates and division probabilities are simultaneously modulated by the delayed differentiated cell population. This allows the model to capture the full spectrum of negative feedback hypothesized experimentally. When all feedback parameters (β_i , γ_{ij}) are set to zero, the model reduces to the basic no-feedback ODE system with constant proliferation and division probabilities.

This hierarchy of models—Type I, Type II and combined—enables systematic investigation of which feedback mechanisms are most important for reproducing the observed CSC and tumor population dynamics.

Experimental Data Used in the Original Study

CSC fraction dynamics in vitro. The experiments track the fraction of CSCs in breast cancer cell populations over time using FACS analysis. Cells are cultured in vitro, and the CSC proportion is measured at multiple time points. Two main scenarios are considered: maintenance of CSC fraction during normal culture conditions (Figure 2A) and regeneration of equilibrium following sorting of CSC populations (Figure 2B). In both cases, the CSC fraction exhibits rapid early changes followed by gradual return toward an equilibrium level.

Relation to simulation figures. The original study compares these experimental observations with simulations of the DDE model. Figures 1C and 1D illustrate simulated tumor population dynamics, while Figures 2A and 2B directly compare simulated CSC fractions to the measured data. Accurate reproduction of these figures requires implementing the DDE model with the reported parameters and plausible initial conditions corresponding to the experimental setups.

Parameter Estimation and Data Fitting

Overview of fitting procedure. To match the model predictions to experimental observations, data points were first extracted from the published figures. The model was then simulated under the reported parameter values, and an objective function was defined to quantify the discrepancy between simulated and observed data. Parameters were estimated using *nonlinear least squares optimization* via MATLAB’s `lsqnonlin` function, allowing us to adjust initial fractions of CSCs and progenitor cells within biologically plausible bounds. Convergence tolerances were set to very low values (10^{-10}) to ensure precise estimation, and the maximum number of iterations and function evaluations were increased to avoid premature termination.

Implementation details. For each model variant—*combined*, *Type I*, *Type II* and *basic*—a dedicated DDE system encodes the dynamics of CSCs, progenitor cells (PCs) and TDCs. Delayed feedback is incorporated either on symmetric division rates (Type II) or proliferation rates (Type I), or both (combined). Simulations were performed using `dde23`, with the initial state defined by the fractions of each cell type scaled to the starting total cell number. The objective function compares the logarithm of the simulated and observed values to reduce sensitivity to large magnitude differences. Optimization returns the best-fitting fractions that minimize the discrepancy between model predictions and experimental data.

Fitting total cell counts versus CSC fractions. When fitting *total cell numbers*, the model residuals are calculated directly from the sum of CSC, PC and TDC compartments. In contrast, when fitting *CSC proportions*, the output is first normalized to compute the fraction

$$\frac{\text{CSC}(t)}{\text{CSC}(t) + \text{PC}(t) + \text{TDC}(t)},$$

and the objective function is evaluated against the measured CSC proportions at the corresponding time points. This distinction ensures that the optimization targets either absolute growth dynamics or relative population composition, which can have markedly different sensitivities to initial conditions and feedback parameters. In both cases, the fitting procedure allows us to systematically explore the parameter space and improve agreement with the experimental trajec-

tories, capturing key features such as transient overshoot or equilibrium levels.

Reproducing Figure 1C: *In vitro*

What Figure 1C demonstrates. Figure 1C of the original paper presents a simulation of tumor cell population dynamics predicted by the DDE model proposed by Liu et al. [2], which describes interactions between CSCs, PCs and TDCs. A central feature of the model is delayed negative feedback from differentiated cells to the CSC compartment, representing regulatory signaling that influences CSC self-renewal. The simulation illustrates how the total tumor cell population evolves over time under the nonlinear growth kinetics implied by this feedback mechanism, demonstrating that CSC-regulated dynamics can produce transient growth patterns before the system approaches a steady state. These dynamics provide a mechanistic explanation for experimentally observed fluctuations in CSC populations during tumor growth. While the authors report the parameter values used in the simulation in the supplementary material, the initial conditions used to generate the trajectories are not specified.

Unspecified initial conditions. Reproducing this figure requires solving a DDE system. Unlike ordinary differential equations, DDE models require both the initial state and a history function over the delay interval. Consequently, the dynamics of the system are sensitive to the choice of initial conditions. Because the original paper does not specify the initial conditions used in the simulations, reproducing the figure requires reconstructing plausible initial states from contextual information in the paper and from the qualitative behavior shown in the figure.

Choice of initial conditions. Because the original paper does not report the initial conditions used in the simulations, reproducing the trajectories requires constructing biologically plausible starting population compositions. In the model of Liu et al. [2], regulation of CSC self-renewal occurs through delayed negative feedback from the differentiated tumor cell compartment. As a result, the initial strength of this feedback depends on the relative abundance of non-CSC cells (PCs and TDCs). Since the early dynamics of the system are sensitive to this feedback term, the initial population composition can significantly influence the resulting trajectory. To explore this sensitivity, two initial condition configurations were selected, representing different regimes of initial feedback strength.

The two initial condition sets considered are defined as proportions of the total tumor cell population and were applied to an initial population of 10^3 cells so that the resulting cell counts fall within the range shown in the published plots.

- **IC 1** — $\text{CSC} = 0.08$, $\text{PC} = 0.07$, $\text{TDC} = 0.85$.

- **IC 2** — $\text{CSC} = 0.015$, $\text{PC} = 0.259$, $\text{TDC} = 0.726$.

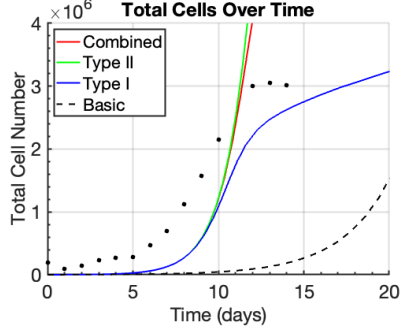
These two configurations represent different initial levels of feedback from the non-CSC compartments. Comparing simulations under these assumptions allows us to assess how sensitive the model dynamics are to the unknown initial population composition and whether the qualitative behavior reported in the original figure is robust to different initial conditions.

Figure comparison and reproduction attempts. Figure 10 compares the original simulation from the paper with several reproduction attempts. The reproduced trajectories were generated using the parameter values reported by the authors and the two initial condition sets described above. As shown in panels (a) and (b), both simulations capture some qualitative aspects of the early growth phase but diverge from the published trajectory at longer time scales. In particular, the reproduced solutions develop pronounced oscillations and overshoot that are not visible in the trajectory shown in the original figure. This discrepancy suggests that the reported parameters and plausible initial conditions alone are insufficient to reproduce the published dynamics, indicating that additional simulation details (such as specific history functions, numerical tolerances or initial delays) may not have been fully specified in the paper.

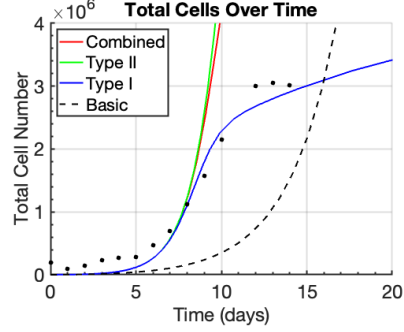
Data-driven fitting and limitations of the model. To further evaluate the model, the data points from the published figure were extracted and used to estimate parameters directly from the observed trajectory. The resulting fit, shown in panel (c) of Figure 10, produces a modest improvement when fitting the *Type I feedback* component alone, while incorporating *Type II feedback* or the *combined feedback* model provides little additional improvement. Despite this adjustment, the model still struggles to accurately match the experimental trend. In particular, the fitted solutions exhibit characteristic *hump-shaped dynamics*, reflecting transient overshoot behavior inherent to the feedback structure of the model. As a result, the model captures the general growth trend and the approximate steady-state level but cannot fully reproduce the smoother, more monotone trajectory observed in the experimental data.

Reproducing Figure 2A: *In vitro*

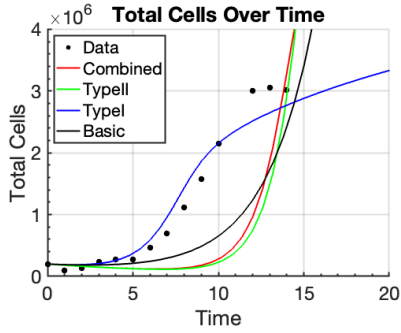
CSC proportion dynamics in vitro. Figure 2A of the original paper compares model predictions with *in vitro* measurements of the CSC fraction over time. In these experiments, cell populations were sorted to obtain an initial CSC proportion and then cultured, after which the CSC fraction was measured at several time points. The model predicts that delayed feedback from differentiated cells regulates CSC self-renewal, allowing the CSC proportion to gradually return toward an equilibrium level. Unlike Figure 1C, which considers total tumor cell counts, this figure focuses on the *relative CSC fraction* within the population.



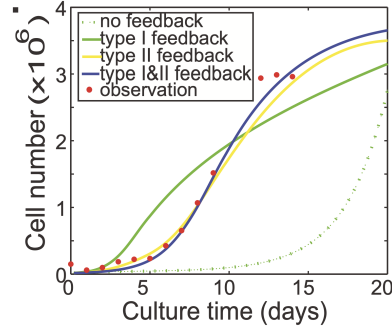
(a) Simulation with reported parameters and initial condition set 1.



(b) Simulation with reported parameters and initial condition set 2.



(c) Model fit to data extracted from the published figure.



(d) Original Figure 1C from the paper.

Figure 10: Comparison between reproduction attempts and the published simulation for Figure 1C. Panels (a) and (b) show simulations generated using the reported parameter values under two different initial condition assumptions. Panel (c) shows the trajectory obtained by fitting the model parameters to data points extracted from the published figure. Panel (d) shows the original simulation reported in the paper.

Reproduction of this figure followed the same approach used for Figure 1C. Simulations were performed using the reported parameter values together with the two initial condition configurations described previously. Because the experimental observations represent CSC *proportions* rather than absolute cell counts, the model output was converted to the CSC fraction before comparison with the extracted data points.

Reproduction attempts and data-driven fitting. For this figure, the initial CSC proportion is fixed by the experimental setup at approximately 1.5%.

Therefore, the initial population composition was set to

$$\text{CSC} = 0.015, \quad \text{PC} = 0.259, \quad \text{TDC} = 0.726.$$

While the proportions are constrained by the experiment, the total number of cells is not specified in the paper. To examine the sensitivity of the model to this quantity, two different initial population sizes were tested:

- **IC 1:** initial population of 10^5 cells with the above proportions.
- **IC 2:** initial population of 10^7 cells with the same proportions.

Although the CSC fraction is identical in both cases, the simulations produce noticeably different trajectories. This occurs because the feedback terms in the model depend on the absolute numbers of non-CSC cells, which alters the effective strength of the delayed regulatory signal. As a result, the total initial population size introduces an additional degree of freedom that significantly influences the predicted CSC dynamics.

For comparison with the experimental measurements, the model output was converted to the CSC fraction and the fitting objective was modified to minimize the discrepancy between the simulated fraction and the experimentally observed CSC proportions at the reported time points. Simulations were performed using the reported parameters and the two initial population sizes described previously. Both simulations capture some qualitative aspects of the CSC recovery dynamics, but significant deviations from the published trajectory remain. Parameter fitting improves agreement noticeably, particularly for the *Type I feedback* model, whereas incorporating *Type II feedback* or the combined feedback model yields only marginal additional improvement.

Observations on feedback dynamics and fitting. The simulations, as shown in Figure 11, reveal several important points about the model behavior. With the smaller initial population (IC 1), all feedback configurations using the reported parameters produce nearly pure exponential decay in the CSC fraction, without any transient overshoot. In contrast, using the larger initial population (IC 2) generates modest *hump-shaped dynamics* for the *Type I* and combined feedback models, reflecting transient overshoot due to delayed regulation, while the *Type II feedback* alone still does not produce humps. Notably, this contrasts with the original publication, which shows hump-shaped trajectories for the *combined* and *Type II feedback* models, highlighting a clear discrepancy between our reproduction attempts and the reported results.

Fitting the model directly to the extracted CSC fraction data significantly improves performance, capturing both the magnitude and timing of the transient overshoot more accurately. Even after fitting, subtle differences remain, likely reflecting the sensitivity of the system to the initial total population size and to unreported modeling or numerical details, such as history functions or

the precise implementation of delayed feedback. These results underscore that the *delayed feedback structure* is capable of producing the characteristic humps observed experimentally, but the observed behavior depends strongly on initial population size and feedback type. Parameter fitting is therefore essential to achieve quantitative agreement and resolve discrepancies between simulations and published trajectories.

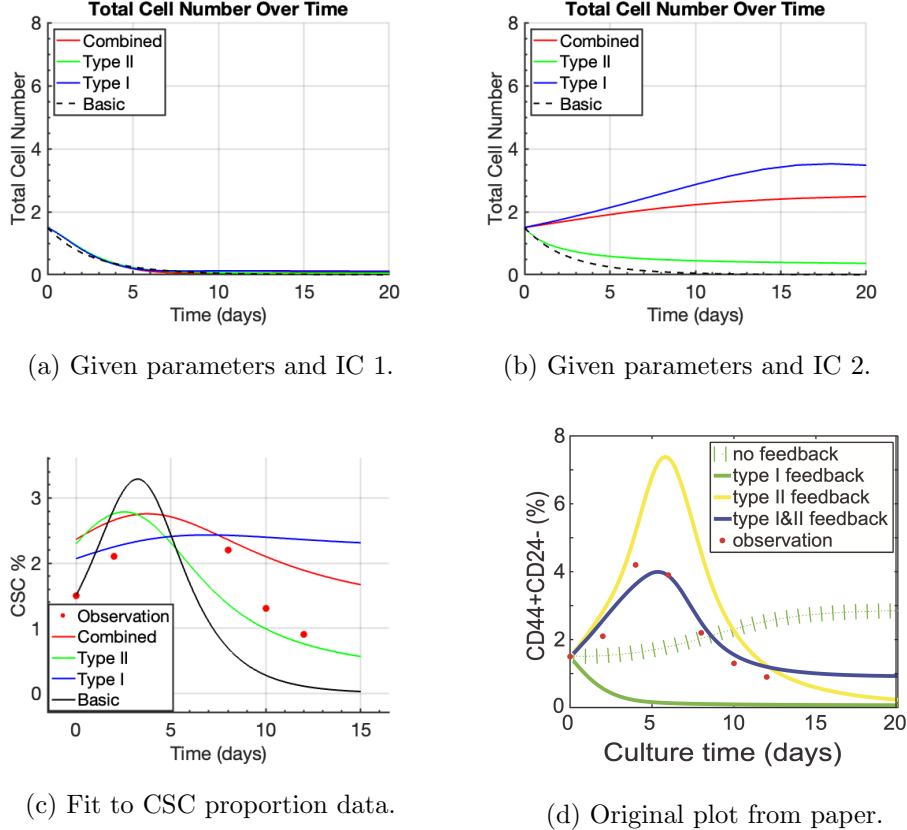


Figure 11: Comparison between reproduction attempts and the published results for Figure 2A. Panels (a) and (b) show simulations using the reported parameters under two initial condition assumptions. Panel (c) shows the trajectory obtained by fitting the model to CSC proportion data extracted from the published figure. Panel (d) shows the original figure from the paper.

Reproducing Figure 1D: *In vivo*

Combined feedback dynamics across datasets. Figure 1D of the original paper compares model predictions with *in vivo* measurements of total tumor cell populations for four datasets: MCF7/HER2 with two replicates and H605 with two replicates. In this figure, the model considers only the *combined feedback*

configuration, which integrates both Type I and Type II delayed regulatory effects on CSC self-renewal. The simulations aim to capture the early transient dynamics as well as longer-term growth behavior observed in the experiments. Unlike Figures 1C and 2A, this figure reports absolute cell counts, so both the initial population composition and total cell numbers influence the dynamics.

Initial conditions and reproduction attempts. To explore plausible initial states, two initial condition sets were tested, chosen to represent low and high initial CSC fractions and correspondingly weak or strong initial negative feedback:

IC 1: CSC = 0.08, PC = 0.07, TDC = 0.85 (total 10^4 cells),

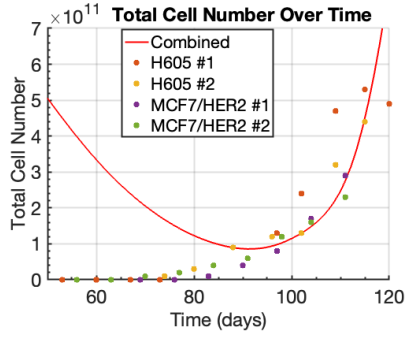
IC 2: CSC = 0.015, PC = 0.259, TDC = 0.726 (total 10^3 cells).

Using the reported parameters, all simulations (see Figure 12) exhibit an initial *hump-shaped* transient with rapid early growth and subsequent decay, reflecting the strong transient response of the combined feedback model. After approximately 50 days, the trajectories attempt to follow the longer-term trends of the experimental data, but deviations remain substantial. Notably, the magnitude and timing of the initial hump are highly sensitive to the choice of initial conditions and total population size.

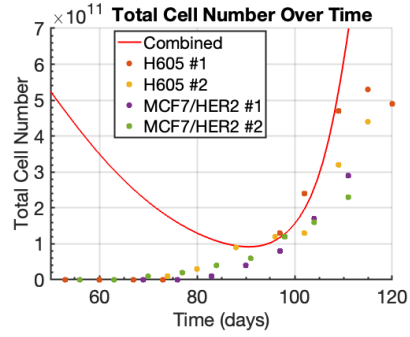
For the manual fits, the model parameters were adjusted to match the experimental trajectories more closely. These fits substantially reduce or eliminate the initial humps, producing smoother trajectories that align better with the observed data. The improvement is particularly pronounced for the H605 datasets, although the MCF7/HER2 replicates also show significant enhancement compared to the original parameter predictions.

Observations on fitting performance and model behavior. Several key points emerge from these simulations. First, the given parameters consistently produce an early hump for both ICs, but the trajectory after the initial peak fails to match the experimental data accurately, especially at later times. Second, adjusting the initial total population influences the transient dynamics, highlighting the importance of reporting absolute cell numbers when reproducing these simulations. Third, the manual fitting procedure greatly improves quantitative agreement, reducing the mismatch and producing trajectories that better follow both the timing and magnitude of growth in all four datasets. Despite this, small discrepancies remain, likely due to unreported numerical or modeling details, such as history functions, precise implementation of the delayed feedback or experimental variability.

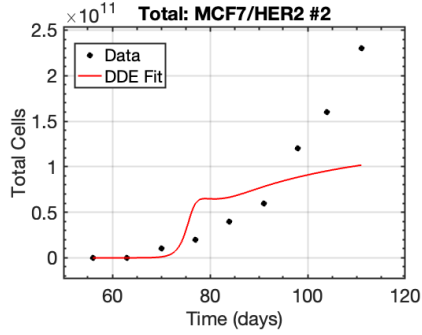
Overall, these results emphasize that the combined feedback model is capable of reproducing the observed growth dynamics, but careful tuning of both initial conditions and parameters is required to resolve the transient overshoot and achieve quantitative agreement with experimental observations.



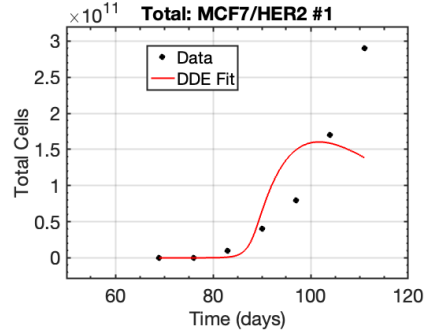
(a) Given parameters with IC 1.



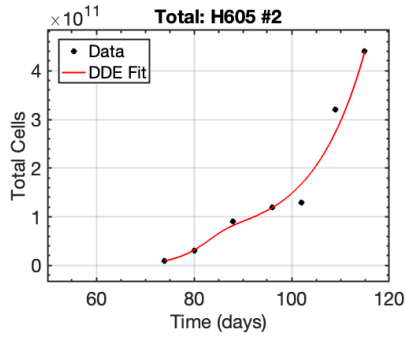
(b) Given parameters with IC 2.



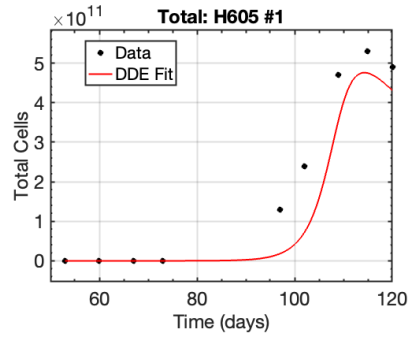
(c) Manual fit, dataset 1.



(d) Manual fit, dataset 2.



(e) Manual fit, dataset 3.



(f) Manual fit, dataset 4.

Figure 12: Reproduction attempts for Figure 1D. Top row: simulations with reported parameters under two initial condition sets. Middle and bottom rows: manual fits to the four experimental datasets, showing improved agreement and reduced transient overshoot compared to the original parameter set.

Reproducing Figure 2B: *In vitro*

CSC fraction dynamics in vitro. Figure 2B of the original paper presents *in vitro* measurements of the CSC fraction over time for a single experimental

dataset. In these experiments, the initial cell population was enriched to a high CSC proportion (approximately 80%), and the CSC fraction was tracked at multiple time points. The model predicts that delayed negative feedback from differentiated cells regulates CSC self-renewal, driving the CSC fraction toward an equilibrium level. Unlike Figures 1C and 2A, this dataset starts with a highly enriched CSC population, which strongly influences early dynamics.

To reproduce this figure, simulations were performed using the reported parameter values and two initial condition configurations with the same CSC-enriched proportions:

$$\text{CSC} = 0.8, \quad \text{PC} = 0.1, \quad \text{TDC} = 0.1.$$

The two configurations differed in total initial cell counts to explore sensitivity to population scale:

- **IC 1** — total population 10^3 cells,
- **IC 2** — total population 10^5 cells.

Because the observations are reported as CSC fractions, the model output was converted to

$$\frac{\text{CSC}(t)}{\text{CSC}(t) + \text{PC}(t) + \text{TDC}(t)}$$

before comparison to the extracted data points.

Reproduction attempts and data-driven fitting. Simulations using the reported parameters fail to reproduce the pronounced hump observed in the published figure, as shown in Figure 13. Both ICs produce nearly pure monotone decay in the CSC fraction. Increasing the total initial population primarily shifts the equilibrium level of the CSC fraction without generating the transient overshoot. In contrast, manually fitting the model to the extracted data produces trajectories that match the data more closely: the CSC fraction initially decreases toward an intermediate level that aligns with the measurements, remains approximately constant for a few time points and then drops to a lower equilibrium. Although the fitted trajectory still does not produce as sharp a hump as implied graphically in the original plot, it provides substantially better agreement with the experimental observations than simulations using the reported parameters.

Observations and discussion. These results highlight several important points about the model and its limitations for high CSC populations. First, the given parameter set is insufficient to reproduce the transient overshoot implied in the published figure, regardless of initial population size. Second, the total number of cells primarily affects the equilibrium level of the CSC fraction rather than the early dynamics, reflecting the dependence of feedback terms on

the absolute number of non-CSC cells. Third, data-driven fitting significantly improves agreement with the observed trajectory, capturing the intermediate plateau and overall trend more accurately. These findings underscore that the *delayed feedback mechanism* in the model can reproduce the general relaxation dynamics of the CSC fraction, but reproducing pronounced early humps may require adjustments to parameter values or additional modeling details not specified in the original publication.

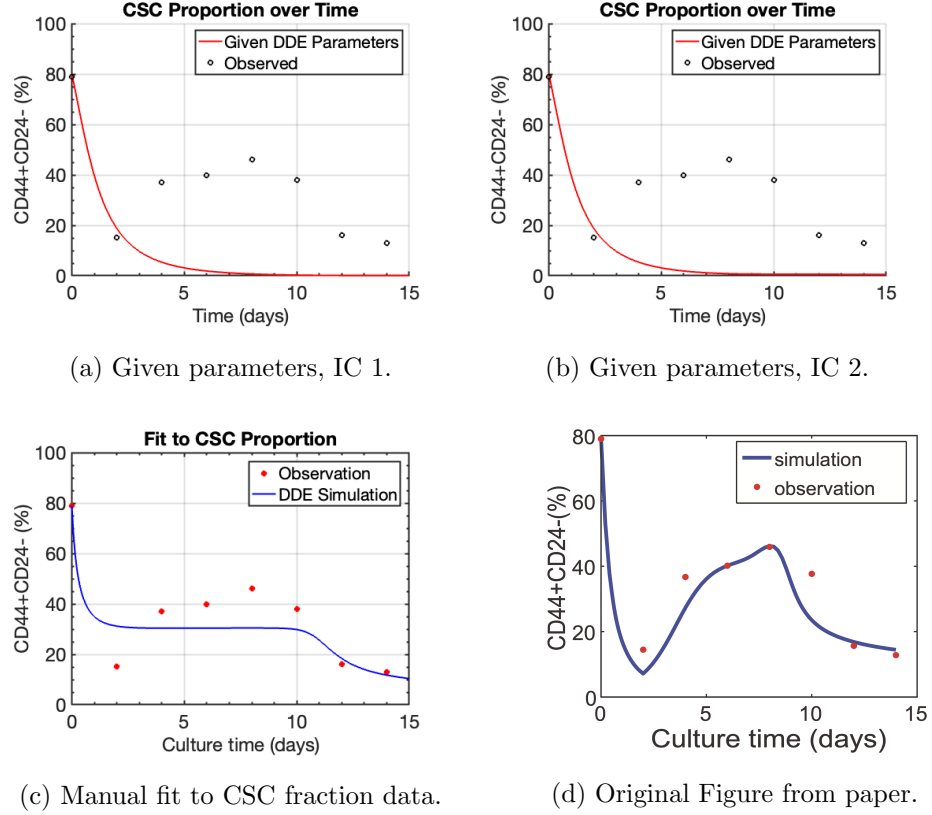


Figure 13: Comparison between reproduction attempts and the published results for Figure 2B. Panels (a) and (b) show simulations using the reported parameters under two different total population assumptions. Panel (c) shows the trajectory obtained by fitting the model to the extracted CSC fraction data. Panel (d) shows the original figure from the paper.

Problem 2: Extra Credit

Use the model in the paper or its variation to fit (or study) one additional set of published experimental data on tumor growth not given in this paper.

Experimental context. To test the generalizability of the CSC-based DDE model, we applied it to an independent tumor growth dataset from the study by Miniére et al. [1]. In this work, the authors evaluated the growth of xenograft tumors in mice under multiple doses of a targeted therapy. Tumor volumes were measured longitudinally at regular intervals using calipers, and the raw measurements were converted into estimated total cell counts assuming a standard cell density for the tumor tissue. This reconstruction provides time-resolved data on the total number of tumor cells, which is compatible with the DDE model framework that tracks hierarchical subpopulations of CSCs, progenitor cells and TDCs.

Rationale for dataset selection. We focused on the 10 nM treatment condition because it represents the lowest dose tested, which exhibits minimal or only delayed effects of the therapy. By restricting the analysis to measurements up to day 13, we capture the intrinsic tumor growth dynamics before the drug exerts substantial influence. During this interval, the tumor growth trajectory exhibits a characteristic “steep–flat–steep” pattern: an initial rapid expansion, followed by a transient plateau likely due to early microenvironmental constraints or feedback effects, and then a renewed increase in cell numbers. This feature makes the dataset especially suitable for evaluating the capacity of feedback-regulated models to reproduce non-monotonic growth dynamics.

Although Miniére et al. [1] do not explicitly measure CSCs or progenitor compartments, the hierarchical compartment model remains relevant. The total tumor expansion reflects the combined contributions of CSCs, progenitors and differentiated cells, and delayed feedback from the differentiated compartment can still modulate the overall growth trajectory. Consequently, even without direct observation of subpopulations, the model can be applied to infer plausible initial CSC and progenitor fractions and to test whether feedback mechanisms can recapitulate the observed growth patterns.

Fitting procedure. We applied the same fitting methodology as in Section 2, adapted for the total tumor cell counts from the six replicates of the 10 nM condition. For each replicate, the initial total cell number was set according to the observed tumor volume at the first time point, while the initial fractions of CSCs and progenitor cells were treated as free parameters to be estimated. The TDC compartment was assigned the remaining fraction to ensure the sum of all compartments equals one.

All six replicates were fitted simultaneously, such that the estimated initial fractions were consistent across replicates, but the total cell number at each time point could vary according to the experimental measurements. The fitting objective function was defined as the sum of squared logarithmic residuals between simulated total cell numbers and the observed tumor counts across all

time points and replicates:

$$\text{Objective} = \sum_{\text{replicates}} \sum_{t_i} [\log N_{\text{sim}}(t_i) - \log N_{\text{obs}}(t_i)]^2,$$

where $N_{\text{sim}}(t_i)$ is the model-predicted total cell number and $N_{\text{obs}}(t_i)$ is the corresponding experimental observation. Using the logarithm ensures that both early small counts and later larger counts contribute proportionally to the fitting.

Optimization was performed using nonlinear least squares (`lsqnonlin`) with bounds restricting initial CSC and progenitor fractions to physically meaningful values between 0 and 1. Standard convergence tolerances were applied, and iterations were monitored to ensure stable parameter estimates. During fitting, all other model parameters, including proliferation rates, division probabilities, death rates and feedback coefficients, were kept fixed at the published values from Liu et al. [2], so that the fitting only adjusted the initial population composition. This approach allows the model to reproduce the overall tumor growth trajectory while preserving the mechanistic feedback structure, even though the individual subpopulations are not directly observed in the dataset.

Results and observations. Fitting results for the six replicates are shown in Figure 14. The model reproduces the overall tumor growth trajectory remarkably well, including the initial steep increase, the transient plateau around intermediate time points and the subsequent acceleration in expansion. These features emerge naturally from the hierarchical compartment structure with delayed feedback, even in the absence of direct CSC or progenitor measurements. By allowing the initial CSC and progenitor fractions to be estimated from the data, the model is able to align with the total cell counts while remaining consistent with biologically plausible population compositions.

Among the model variants, the *Type II* and *combined* feedback models produce the closest fits across replicates. This is expected given that the early plateau in the growth curves corresponds to a temporary suppression of effective self-renewal, which is directly captured by Type II feedback on symmetric and asymmetric division probabilities. The *Type I* feedback model, which modulates only proliferation rates, provides a reasonably close fit but fails to capture subtle aspects of the plateau as precisely, highlighting the importance of feedback acting on self-renewal in this system.

Replicate-to-replicate variability is modest and largely reflected in differences in the estimated initial CSC fractions, suggesting that the primary source of heterogeneity in tumor growth can be explained by differences in the starting composition of the hierarchical cell population. The approach of fitting only the initial fractions while keeping other parameters fixed demonstrates that the model structure itself is sufficient to explain non-monotonic growth dynamics, providing a robust test of the feedback hypotheses without overfitting.

Broader observations indicate that compartmental DDE models calibrated

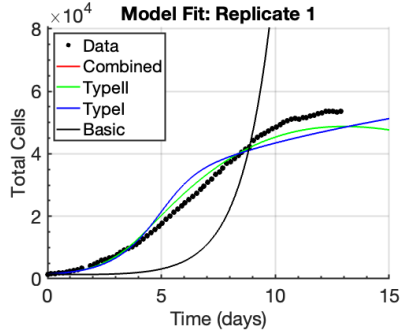
to total tumor counts can successfully infer hidden stem and progenitor dynamics, even when experimental measurements do not resolve subpopulations. The high-quality fits reinforce the biological plausibility of delayed feedback in regulating early tumor growth, and they validate the approach of using minimal free parameters (here, initial fractions) to capture essential features of the dynamics. Overall, these results support the generalizability of the CSC-based feedback model to independent experimental datasets and highlight its utility for interpreting complex growth patterns in hierarchical tumor populations.

Code availability

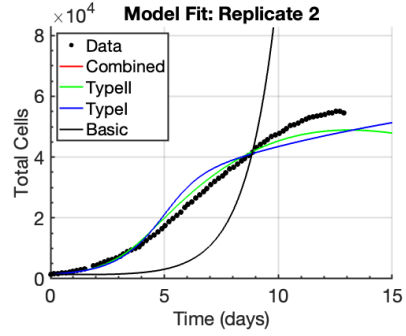
All the code used for these problems is available publicly at this GitHub repository.

References

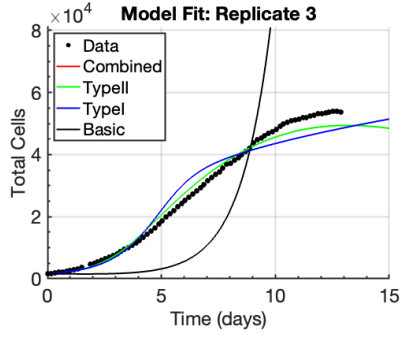
- [1] Minière, H. J. M., Lima, E. A. B., Lorenzo, G., Hormuth II, D. A., Ty, S., Brock, A., & Yankeelov, T. E. (2024). A mathematical model for predicting the spatiotemporal response of breast cancer cells treated with doxorubicin. *Cancer Biology & Therapy*, 25(1), 2321769.
- [2] Liu, X., Johnson, S., Liu, S., Kanojia, D., Yue, W., Singh, U. P., Wang, Q., Wang, Q., Nie, Q., & Chen, H. (2013). Nonlinear growth kinetics of breast cancer stem cells: implications for cancer stem cell targeted therapy. *Scientific Reports*, 3(1), 2473.



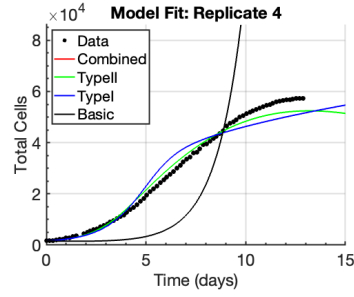
(a) Fit for replicate 1.



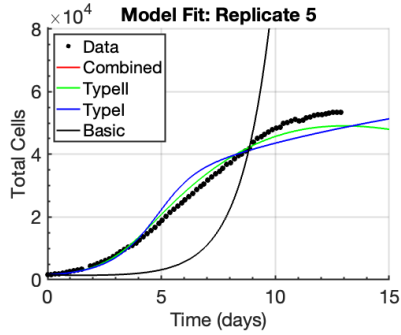
(b) Fit for replicate 2.



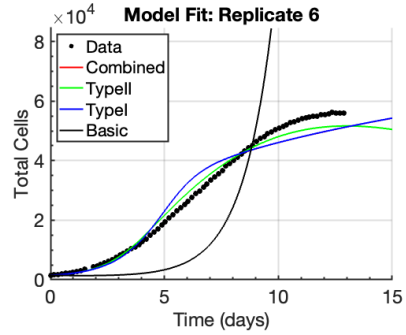
(c) Fit for replicate 3.



(d) Fit for replicate 4.



(e) Fit for replicate 5.



(f) Fit for replicate 6.

Figure 14: Fitting the CSC-based DDE model to six replicates of tumor growth under the 10 nM condition from Miniére et al. [1]. Each panel (a–f) shows the model fit to one replicate, capturing the early rise, transient plateau and subsequent expansion. Type II and combined feedback models provide the closest agreement with the observed growth trajectories. **The red curve for the combined model closely tracks the green curve for Type II, and in several panels is visually indistinguishable.**